

Posterior Collapse as a Phase Transition in Variational Autoencoders

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Abstract

We investigate the phenomenon of posterior collapse in variational autoencoders (VAEs) from the perspective of statistical physics, and reveal that it constitutes a phase transition governed jointly by data structure and model hyper-parameters. By analyzing the stability of the trivial solution associated with posterior collapse, we identify a critical hyper-parameter threshold. This critical boundary, separating meaningful latent inference from collapse, is characterized by a discontinuity in the KL divergence between the approximate posterior and the prior distribution. We validate this critical behavior on both synthetic and real-world datasets, confirming the existence of a phase transition. Our results demonstrate that posterior collapse is not merely an optimization failure, but rather an emerging phase transition arising from the interplay between data structure and variational constraints. This perspective offers new insights into the trainability and representational capacity of deep generative models.

Keywords: Variational autoencoders, Posterior collapse, Phase transition

1. Introduction

Variational autoencoders (VAEs) are a class of probabilistic generative models that integrate variational inference with deep learning architectures [1, 2, 3]. Designed to learn latent representations of high-dimensional data, VAEs approximate the intractable posterior distribution over latent variables by introducing a parametric variational family, typically realized via neural networks. The learning objective is to maximize the evidence lower bound (ELBO), which balances reconstruction accuracy and the complexity of the latent encoding through a regularization term given by the Kullback–Leibler (KL) divergence between the approximate posterior and a prior distribution [1, 2, 3, 4].

VAEs have been widely and successfully applied in tasks such as image synthesis [5], anomaly detection [6], and molecular design [7]. However, as a data driven method, it is pointed out that the performance of VAEs is highly effected by data itself [8, 9, 10]. Moreover, VAEs often suffer from a phenomenon known as posterior collapse, in which a subset of the latent variables is effectively ignored [11, 12, 13, 14]. It is commonly pointed that posterior collapse reasons from the KL-divergence term in the ELBO, which encourages the approximate posterior distribution to match the prior distribution for latent variables [15, 16, 17, 18]. Several approaches have been proposed to address this issue, such as introducing a weighting coefficient β in the ELBO (β -VAE) [19], applying anneal methods [18], or rewriting ELBO itself [20].

Recent works have provided new insights into the causes of posterior collapse by studying linear VAE models [11, 12, 13, 14]. These studies highlight alternative contributing factors, such as the the log marginal likelihood [11], the structure of the posterior distribution [12], competition among terms in the ELBO [13], and bad local extrema [14], are also responsible for posterior collapse.

In this work, we present a new perspective on posterior collapse in VAEs by framing it as a phase transition [21]. In section 3, we revisit the mathematical formulation of VAEs and derive conditions for maximizing the ELBO. Section

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4 introduces our key insight: posterior collapse can be interpreted as a phase transition in VAEs, characterized by the emergence of a critical point. We validate this view using both synthetic and real-world datasets. Finally, section 5 presents a discussion and concluding remarks.

2. Related works

Bayesian deep learning: VAEs represent a central approach in Bayesian deep learning, combining deep neural networks with variational inference to approximate intractable posteriors [1, 3]. The ELBO objective provides a principled way to learn latent-variable generative models by balancing reconstruction fidelity and latent regularization. Subsequent developments such as β -VAE [19] and ladder VAE [17] further explore trade-offs between disentanglement, expressivity, and inference accuracy. Recent surveys [2] have highlighted the growing diversity of Bayesian generative models, including extensions to hierarchical and dynamic settings.

Posterior collapse: Despite their flexibility, VAEs are known to suffer from posterior collapse, where the approximate posterior becomes indistinguishable from the prior, rendering latent variables uninformative [15, 11, 14]. Early explanations attributed this issue to the strong regularizing effect of the KL-divergence in the ELBO, motivating strategies such as KL annealing and β -VAE modifications [18, 19]. Later work examined additional causes, including poor decoder architecture, inference mismatch, and suboptimal optimization dynamics [22, 20]. More recently, researchers have proposed that posterior collapse can result from deeper statistical properties of the learning objective itself [11, 12, 13], hinting at the need for a more principled theoretical framework.

Linear models and analytical perspectives: To better understand the mechanism of posterior collapse, several works have studied VAEs in linear settings where analytical insights are tractable [11, 12, 13]. These studies reveal that collapse can occur even in the absence of optimization failures, due to properties intrinsic to the ELBO landscape—such as the dominance of the log marginal likelihood or the flatness of local extrema. These findings have motivated our approach, which extends beyond linear assumptions by analyzing posterior collapse as a phase transition determined jointly by data structure and model hyper-parameters.

3. Maximizing evidence lower bound

A VAE compresses high-dimensional data $\mathbf{x} \in \mathbb{R}^N$ into low-dimensional latent variables $\mathbf{z} \in \mathbb{R}^n$ (with $N > n$) using a prior distribution $p_{\text{lat}}(\mathbf{z})$ via an encoder $q_\phi(\mathbf{z}|\mathbf{x})$, and generate data from latent variables via a decoder $p_\theta(\mathbf{x}|\mathbf{z})$ [1, 2]. Due to the fact that the distribution $p_{\text{data}}(\mathbf{x})$ for data $\mathbf{x}$ is usually intractable, the encoder $q_\phi(\mathbf{z}|\mathbf{x})$, also known as variational posterior distribution, approximates the posterior distribution $p(\mathbf{z}|\mathbf{x})$ via amortized variational inference [12, 3], which is tractable using the evidence lower bound (ELBO) [1]:

$$\text{ELBO}(\mathbf{x}) \equiv \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\ln p_\theta(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_{\text{lat}}(\mathbf{z})) \leq \ln p_{\text{data}}(\mathbf{x}). \quad (1)$$

Training a VAE aims to maximize the ELBO, which includes the reconstruction loss $\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\ln p_\theta(\mathbf{x}|\mathbf{z})]$ and the KL-divergence $D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_{\text{lat}}(\mathbf{z}))$ as regularization [1, 12]. As large amount of data $\mathbf{x}$ are used for training, the target for a VAE can be also regarded as the expectation of ELBO [23, 13]:

$$\begin{aligned} \mathcal{L}_{\text{VAE}} &\equiv \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\text{ELBO}(\mathbf{x})] \\ &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left[\mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} [\ln p_\theta(\mathbf{x}|\mathbf{z})] \right] - \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left[D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_{\text{lat}}(\mathbf{z})) \right] \\ &= \int \ln p_\theta(\mathbf{x}|\mathbf{z}) q_\phi(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} d\mathbf{z} - \int q_\phi(\mathbf{z}|\mathbf{x}) \ln \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_{\text{lat}}(\mathbf{z})} p_{\text{data}}(\mathbf{x}) d\mathbf{x} d\mathbf{z}. \end{aligned} \quad (2)$$

The extrema of $\mathcal{L}_{\text{VAE}}$ can be obtained by performing variational calculus on

$$\mathcal{I}[p_\theta(\mathbf{x}|\mathbf{z}), q_\phi(\mathbf{z}|\mathbf{x})] \equiv \mathcal{L}_{\text{VAE}} - \lambda_1(\mathbf{z}) \left[\int p_\theta(\mathbf{x}|\mathbf{z}) d\mathbf{x} - 1 \right] - \lambda_2(\mathbf{x}) \left[\int q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z} - 1 \right], \quad (3)$$

where Lagrange multipliers $\lambda_1(\mathbf{z})$ and $\lambda_2(\mathbf{x})$ are introduced for the normalizations

$$\int p_\theta(\mathbf{x}|\mathbf{z})d\mathbf{x} = 1, \quad (4)$$

$$\int q_\phi(\mathbf{z}|\mathbf{x})d\mathbf{z} = 1, \quad (5)$$

respectively. The extrema points reads:

$$p_\theta(\mathbf{x}|\mathbf{z}) = p(\mathbf{x}|\mathbf{z}) \quad (6)$$

$$q_\phi(\mathbf{z}|\mathbf{x}) = p(\mathbf{z}|\mathbf{x}), \quad (7)$$

where the multipliers are

$$\lambda_1(\mathbf{z}) = p_{\text{lat}}(\mathbf{z}) \quad (8)$$

$$\lambda_2(\mathbf{x}) = p_{\text{data}}(\mathbf{x}) [\ln p_{\text{data}}(\mathbf{x}) - 1]. \quad (9)$$

The meaning of the extrema is intuitive: the conditional distributions $p(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z}|\mathbf{x})$ are required to perfectly encode and reconstruct the data from the latent variables. In this case, it is easy to check the extrema value

$$\mathcal{L}_{\text{VAE}}^* = \int p_{\text{data}}(\mathbf{x}) \ln p_{\text{data}}(\mathbf{x})d\mathbf{x}, \quad (10)$$

which is the log likelihood, or the opposite of the entropy, of data.

It is worth noting that there are multiple extrema points, as there is more than one set of conditional distributions. The existence of multiple extrema of the ELBO has also been noted in previous works, such as [24, 25]. We can also understand it by analyzing the stability of extrema points. By applying second order variation on Eq. (3) and substitute Eqs. (6) and (7), we have the Hessian matrix $\mathbf{H}(\mathbf{x}, \mathbf{y})$ for the extrema points as:

$$\mathbf{H}(\mathbf{x}, \mathbf{y}) = \left(\begin{array}{cc} \frac{\delta^2 \mathcal{I}}{\delta p_\theta^2} & \frac{\delta^2 \mathcal{I}}{\delta p_\theta \delta q_\phi} \\ \frac{\delta^2 \mathcal{I}}{\delta p_\theta \delta q_\phi} & \frac{\delta^2 \mathcal{I}}{\delta q_\phi^2} \end{array} \right) \bigg|_{\substack{p_\theta(\mathbf{x}|\mathbf{z})=p(\mathbf{x}|\mathbf{z}) \\ q_\phi(\mathbf{z}|\mathbf{x})=p(\mathbf{z}|\mathbf{x})}} = \left(\begin{array}{cc} -\frac{p(\mathbf{z}|\mathbf{x})}{p^2(\mathbf{x}|\mathbf{z})} & \frac{1}{p(\mathbf{x}|\mathbf{z})} \\ \frac{1}{p(\mathbf{x}|\mathbf{z})} & -\frac{1}{q(\mathbf{z}|\mathbf{x})} \end{array} \right) p_{\text{data}}(\mathbf{x}), \quad (11)$$

which is negative semi-definite, implying the flatness along certain directions around extrema points. This further implies the existence of multiple extrema, as illustrated in Figure 1.

There is a special extrema point,

$$p_\theta(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}), \quad (12)$$

$$q_\phi(\mathbf{z}|\mathbf{x}) = p_{\text{lat}}(\mathbf{z}), \quad (13)$$

known as posterior collapse (also in Figure 1), where the variational posterior distribution $q_\phi(\mathbf{z}|\mathbf{x})$ converges to the prior distribution $p_{\text{lat}}(\mathbf{z})$, rendering the latent variables uninformative about the data, and limiting the capacity of VAEs to extract and represent meaningful structure from the data [13, 15, 12, 11].

In practice, directly approximating the true posterior distribution is challenging. The standard approach is to assume specific forms for $p_\theta(\mathbf{x}|\mathbf{z})$ and $q_\phi(\mathbf{z}|\mathbf{x})$, and to approximate parameters such as the mean and variance using deep learning. One of the commonly chosen form reads:

$$p_\theta(\mathbf{x}|\mathbf{z}) \equiv \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma'} \exp \left\{ -\frac{[x_i - f_i(\mathbf{z})]^2}{2\sigma'^2} \right\}, \quad (14)$$

$$q_\phi(\mathbf{z}|\mathbf{x}) \equiv \prod_{j=1}^n \frac{1}{\sqrt{2\pi}\sigma_j(\mathbf{x})} \exp \left\{ -\frac{[z_j - \mu_j(\mathbf{x})]^2}{2\sigma_j^2(\mathbf{x})} \right\}, \quad (15)$$

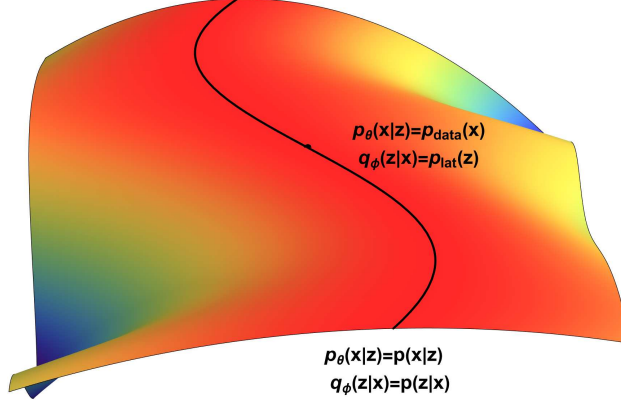


Figure 1: Sketch of the $\mathcal{L}_{\text{VAE}}$ surface. There exists multiple extrema points (black curve), around which the surface is flat along certain directions. Posterior collapse $p_\theta(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x})$, $q_\phi(\mathbf{z}|\mathbf{x}) = p_{\text{lat}}(\mathbf{z})$ (black dot) is also an extrema point.

while the prior distribution

$$p_{\text{lat}}(\mathbf{z}) \equiv \prod_{j=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z_j^2}{2}\right), \quad (16)$$

is standard Gaussian. This form is known as deep Gaussian VAEs, as both $p_\theta(\mathbf{x}|\mathbf{z})$ and $q_\phi(\mathbf{z}|\mathbf{x})$ are Gaussian with the parameters $\mathbf{f}(\mathbf{z})$, $\boldsymbol{\mu}(\mathbf{x})$ and $\boldsymbol{\sigma}(\mathbf{x})$ learned from networks [11, 23]. The hyper-parameter σ'^2 describes the reconstruction noise variance.

Substitute Eqs. (14)–(16) into Eq. (2), we have the training target for deep Gaussian VAEs [23, 11, 13]:

$$\begin{aligned} \mathcal{L}_{\text{VAE}}^G &\equiv N \ln \frac{1}{\sqrt{2\pi}\sigma'} - \int \frac{1}{2\sigma'^2} \sum_{i=1}^N [x_i - f_i(\mathbf{z})]^2 q_\phi(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} d\mathbf{z} \\ &\quad - \frac{1}{2} \int \sum_{j=1}^n [\sigma_j^2(\mathbf{x}) + \mu_j^2(\mathbf{x}) - 1 - \ln \sigma_j^2(\mathbf{x})] p_{\text{data}}(\mathbf{x}) d\mathbf{x}. \end{aligned} \quad (17)$$

By applying variation on Eq. (17), we derive the conditions for extrema points:

$$f_i(\mathbf{z}) = \frac{\int x_i q_\phi(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}}{\int q_\phi(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}}, \quad (18)$$

$$\mu_j(\mathbf{x}) = \frac{\frac{1}{2\sigma'^2} \int \sum_{i=1}^N [x_i - f_i(\mathbf{z})]^2 z_j q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z}}{\int \sum_{i=1}^N [x_i - f_i(\mathbf{z})]^2 q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z} - \sigma_j^2(\mathbf{x})}, \quad (19)$$

and

$$\sigma_j^4(\mathbf{x}) - \left\{ 1 + \frac{1}{2\sigma'^2} \int \sum_{i=1}^N [x_i - f_i(\mathbf{z})]^2 q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z} \right\} \sigma_j^2(\mathbf{x}) + \frac{1}{2\sigma'^2} \int \sum_{i=1}^N [x_i - f_i(\mathbf{z})]^2 [z_j - \mu_j(\mathbf{z})]^2 q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z} = 0. \quad (20)$$

It is easy to find a trivial extrema point:

$$f_i(\mathbf{z}) = \langle x_i \rangle, \quad (21)$$

$$\mu_j(\mathbf{x}) = 0, \quad (22)$$

$$\sigma_j(\mathbf{x}) = 1, \quad (23)$$

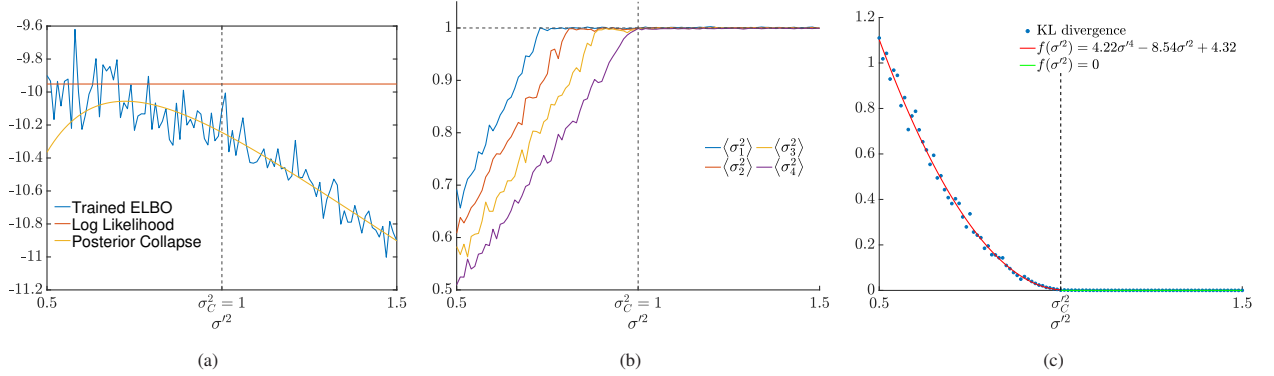


Figure 2: (a) Trained ELBO, log likelihood and posterior collapse line v.s. σ^2 . The trained ELBO converges the posterior collapse when $\sigma^2 > \sigma_C^2 = 1$. (b) Mean latent variances $\langle \sigma_j^2 \rangle$ ($j = 1, 2, 3, 4$) v.s. σ^2 . All of them equal 1 when $\sigma^2 > \sigma_C^2 = 1$. (c) KL divergence v.s. σ^2 , which converges to 0 when $\sigma^2 > \sigma_C^2 = 1$ (green line), and can be fitted by a quadratic function of σ^2 when $\sigma^2 < \sigma_C^2 = 1$ (red line).

which corresponds to posterior collapse, as $q_\phi(\mathbf{z}|\mathbf{x}) = p_{\text{lat}}(\mathbf{z})$ at this point. Here, we denote $\langle \theta \rangle \equiv \mathbb{E}_{p_{\text{data}}(\mathbf{x})}[\theta]$. We can also find the extrema value for posterior collapse,

$$\mathcal{L}_{\text{VAE}}^{\text{G}*} = -\frac{N}{2} \ln 2\pi\sigma'^2 - \frac{1}{2\sigma'^2} \text{Tr}(\mathbf{\Sigma}), \quad (24)$$

where $\mathbf{\Sigma}$ is the covariance matrix for data $\mathbf{x}$. It should be noted that this extremum, which also depends on the hyper-parameter σ'^2 , generally does not coincide with the data's log-likelihood.

4. Posterior collapse as phase transition

Previous studies on posterior collapse have primarily focused on linear VAE models [11, 12, 13], which adopt $p_\theta(\mathbf{x}|\mathbf{z})$, $q_\phi(\mathbf{z}|\mathbf{x})$ and $p_{\text{lat}}(\mathbf{z})$ as defined in Eqs. (14)–(16), with both $\mathbf{f}(\mathbf{z})$ and $\boldsymbol{\mu}(\mathbf{x})$ taken as linear functions. Here, we aim to provide an interpretation of posterior collapse as a phase transition, with no limitations on the forms of $\mathbf{f}(\mathbf{z})$, $\boldsymbol{\mu}(\mathbf{x})$ and $\boldsymbol{\sigma}(\mathbf{x})$.

As what have discussed before, posterior collapse $q_\phi(\mathbf{z}|\mathbf{x}) = p_{\text{lat}}(\mathbf{z})$ is a trivial solution for VAEs. Moreover, if we assume the form of $q_\phi(\mathbf{z}|\mathbf{x})$, such as the choice in Eq. (15), it is possible that the equation

$$p_{\text{lat}}(\mathbf{z}) = \int q_\phi(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}, \quad (25)$$

has no solution other than $q_\phi(\mathbf{z}|\mathbf{x}) = p_{\text{lat}}(\mathbf{z})$ under some $p_{\text{data}}(\mathbf{x})$ [22]. Thus, it is reasonable to regard the onset of posterior collapse as a phase transition: from nontrivial, possible multiple extrema points to the trivial one. Another evidence for claiming posterior collapse as phase transition falls on the KL-divergence term in ELBO. The KL-divergence $D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_{\text{lat}}(\mathbf{z}))$ becomes and keeps 0 if fallen into posterior collapse, which may lead to discontinuities in the derivative of the KL divergence.

We propose a simple criterion to approximately determine whether posterior collapse occurs for VAEs with the settings as Eqs. (14)–(16):

$$\sigma'^2 > \max[\xi_1^2, \dots, \xi_N^2], \quad (26)$$

where ξ_i^2 ($i = 1, 2, \dots, N$) are the eigenvalues of the covariance matrix $\mathbf{\Sigma}$ for the data. We proof this criterion in the Appendix.

We validate the criterion (26) using both synthetic and real-world datasets. For the synthetic dataset, we use 256,000 samples of 8-dimensional data ($N = 8$) drawn from a zero-mean Gaussian distribution, with eigenvalues of the covariance matrix $\mathbf{\Sigma}$ given by $\xi_i^2 = \exp\{-(i-1)\lambda\}$ for $i = 1, 2, \dots, N$. We apply the VAE model with two hidden layers in both the encoder (for learning $\boldsymbol{\mu}(\mathbf{x})$ and $\boldsymbol{\sigma}(\mathbf{x})$ in $q_\phi(\mathbf{z}|\mathbf{x})$) and the decoder (for learning $\mathbf{f}(\mathbf{z})$ for $p_\theta(\mathbf{x}|\mathbf{z})$). The

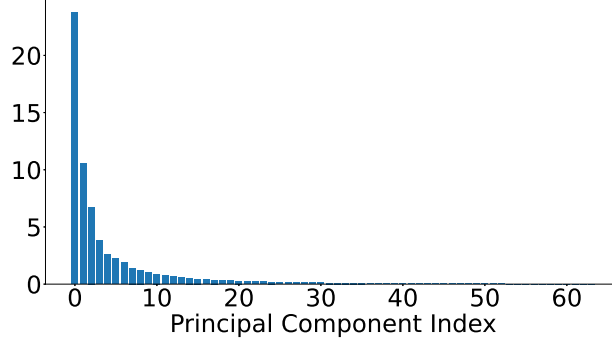


Figure 3: Principle components of the preprocessed CIFAR10 data. The maximum one is 23.78, which corresponds to $\sigma_C'^2$.

encoder has hidden layers of dimensions 256 and 128 with ReLU activations, while the decoder has hidden layers of dimensions 128 and 256, also with ReLU activations. We train and record the trained ELBO, KL-divergence, log likelihood, posterior collapse (Eq. (24)), and mean latent variances $\langle \sigma_j^2 \rangle$ ($j = 1, 2, \dots, n$).

We set $\lambda = 0.1$, $n = 4$, and vary σ'^2 . The critical point is $\sigma_C'^2 = 1$. The trained ELBO converges to the posterior collapse line (Figure. 2a), and $\langle \sigma_j^2 \rangle = 1$ when $\sigma'^2 \geq \sigma_C'^2$ (Figure 2b). Furthermore, the KL divergence approaches 0 when $\sigma'^2 \geq \sigma_C'^2$ (Figure 2c). We also observe a discontinuity in the derivatives of the KL divergence at $\sigma'^2 = \sigma_C'^2$ by fitting the KL divergence with a piecewise function

$$f(\sigma'^2) = \begin{cases} 4.22\sigma'^4 - 8.54\sigma'^2 + 4.32 & (\sigma'^2 < \sigma_C'^2 = 1) \\ 0 & (\sigma'^2 > \sigma_C'^2 = 1) \end{cases}, \quad (27)$$

of which the first-order derivative

$$f'(\sigma'^2) = \begin{cases} 8.44\sigma'^2 - 8.54 & (\sigma'^2 < \sigma_C'^2 = 1) \\ 0 & (\sigma'^2 > \sigma_C'^2 = 1) \end{cases}, \quad (28)$$

and second-order derivative

$$f''(\sigma'^2) = \begin{cases} 8.44 & (\sigma'^2 < \sigma_C'^2 = 1) \\ 0 & (\sigma'^2 > \sigma_C'^2 = 1) \end{cases}, \quad (29)$$

show discontinuous. This discontinuity is indicative of a phase transition in the VAE.

We choose CIFAR10 [26] as real world data. CIFAR-10 is a widely used image dataset consisting of 60,000 color images with a resolution of 32×32 . To speed up training, we preprocess the dataset by converting color images to grayscale, resizing them to 8×8 pixels ($N = 64$), and applying standard normalization:

$$x_i \rightarrow x'_i = \frac{x_i - \langle x_i \rangle}{\sqrt{\langle x_i^2 \rangle - \langle x_i \rangle^2}}. \quad (30)$$

The principle components of $\mathbf{x}'$ are shown in Figure. 3, with the largest being 23.78, implying $\sigma_C'^2 \approx 23.78$. The architecture of the VAE model is similar to that used for the synthetic dataset, but the dimensions of hidden layers are 512 and 256 for the encoder, while 256 and 512 for the decoder. We set $n = 8$ for latent variables. Similar behaviors are observed near the critical point $\sigma_C'^2 \approx 23.78$, as seen in the ELBO (Figure. 4a), the mean latent variances $\langle \sigma_j^2 \rangle$ (Figure. 4b) and the KL divergence (Figure. 4c). We can also explore the discontinuous of KL-divergence by fitting it with the following piecewise function (red line and green line in Figure. 4c):

$$f(\sigma'^2) = \begin{cases} 6.36(\sigma'^2)^{-0.71} - 0.71 & (\sigma'^2 < \sigma_C'^2 \approx 23.78) \\ 0 & (\sigma'^2 > \sigma_C'^2 \approx 23.78) \end{cases}, \quad (31)$$

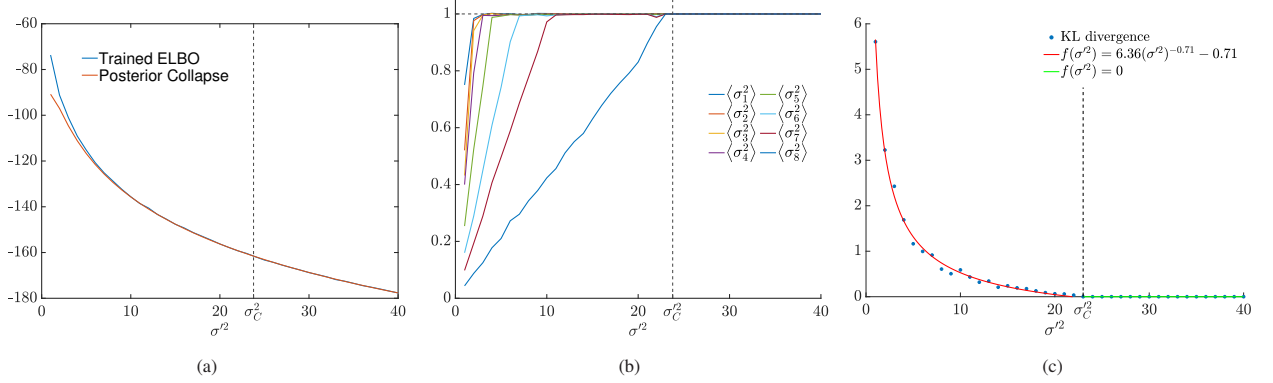


Figure 4: (a) Trained ELBO and posterior collapse line v.s. σ'^2 . The trained ELBO converges the posterior collapse when $\sigma'^2 > \sigma_C'^2 \approx 23.78$. (b) Mean latent variances $\langle \sigma_j'^2 \rangle$ ($j = 1, \dots, 8$) v.s. σ'^2 . All of them equal 1 when $\sigma'^2 > \sigma_C'^2 \approx 23.78$. (c) KL divergence v.s. σ'^2 , which converges to 0 when $\sigma'^2 > \sigma_C'^2 \approx 23.78$.

5. Conclusions

In this work, we have established a new perspective on posterior collapse in VAEs by interpreting it as a phase transition driven by the interplay between data structure and model hyper-parameters. By analyzing the stability of the trivial solution corresponding to posterior collapse, we derived a critical point characterized by the spectral properties of the data covariance. This condition predicts the onset of collapse, which we validated on both constructed and real world datasets.

Our findings suggest that posterior collapse is not merely an artifact of optimization dynamics but a fundamental emergent behavior governed by statistical and variational constraints. This perspective not only bridges variational learning with phase transition theory but also offers practical tools for preventing posterior collapse in VAE training.

Importantly, the theoretical criterion we derive is consistent with prior results in linear VAEs [11, 12, 13], as the stability of the collapse point is mainly determined by linear perturbations of the encoder and decoder functions. Moreover, beyond the bad local optima [14], our analysis offers a complementary perspective by identifying posterior collapse as a kind of phase transition that reflects global data-model interactions.

Further, the role of the reconstruction noise variance σ'^2 is analogous to the β parameter in β -VAE, adjusting the weights of the reconstruction loss and KL-divergence in the ELBO. As such, our framework can be readily extended to β -VAEs, since introducing a β coefficient for the KL-divergence is equivalent to rescaling the ELBO and σ'^2 , and shifting the critical point accordingly.

Future work could extend the current analysis to settings with non-Gaussian data distributions or decoders, where the reconstruction loss no longer has a simple quadratic form. Such extensions could reveal new types of phase behavior and potentially uncover richer collapse dynamics influenced by the shape and tails of the data distribution. In addition, it is worth investigating the generality of this phase transition framework in extended architectures such as conditional or hierarchical VAEs, and further explore its implications in understanding the expressivity and limitations of deep generative models.

Moreover, it is also worth exploring more precious theories to identify the posterior collapse as, and analysis the order of the phase transition, beyond discussing the stability of the trivial solutions (21)–(23). Such theories may require analytical solutions of extremes other than the trivial one, and the corresponding stability.

Another promising direction is to investigate how posterior collapse emerges dynamically during training. As suggested by Figures 2b and 4b, and previous works on linear models [11, 12, 13], the collapse of the variational posterior may not occur simultaneously across all latent dimensions, but rather in a sequential manner. Specifically, each latent variance $\langle \sigma_j'^2 \rangle$ appears to approach 1 at different choice of the hyper-parameter σ'^2 , implying that posterior collapse may form via the successive destabilization of individual latent directions. This observation raises the possibility of a more nuanced collapse mechanism governed by the alignment of latent directions with the principal components of the data. Developing a refined theoretical framework that accounts for this dimension-wise collapse

could provide deeper insight into the onset of collapse and suggest new strategies—such as targeted regularization or selective annealing—to mitigate it and preserve informative latent structure.

6. Acknowledgements

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Appendix A. Derivation criterion (26)

Criterion (26) comes from analyzing the stability of the trivial extremum given by (21)–(23). If the trivial local extrema of $\mathcal{L}_{\text{VAE}}^G$ is unstable, the VAE may escape posterior collapse; otherwise, it may converge to the collapsed solution. We analyze this stability with the following framework.

The second order variation of $\mathcal{L}_{\text{VAE}}^G$ in Eq. (17) at the trivial extrema point (21)–(23) reads:

$$\delta^2 \mathcal{L}_{\text{VAE}}^G = \int (\delta f, \delta \mu, \delta \sigma) \mathbf{H}^G (\delta f, \delta \mu, \delta \sigma)^\top d\tilde{\mathbf{x}} d\mathbf{z}, \quad (\text{A.1})$$

where

$$\mathbf{H}^G = \begin{pmatrix} \mathbf{A} & \mathbf{D} & \mathbf{E} \\ \mathbf{D}^\top & \mathbf{B} & \mathbf{0} \\ \mathbf{E}^\top & \mathbf{0} & \mathbf{C} \end{pmatrix}, \quad (\text{A.2})$$

$$A_{ij} = -\frac{\delta_{ij}}{\sigma^2} \tilde{p}_{\text{data}}(\tilde{\mathbf{x}}) p_{\text{lat}}(\mathbf{z}), \quad (\text{A.3})$$

$$B_{ij} = -\delta_{ij} \tilde{p}_{\text{data}}(\tilde{\mathbf{x}}) p_{\text{lat}}(\mathbf{z}), \quad (\text{A.4})$$

$$C_{ij} = -2\delta_{ij} \tilde{p}_{\text{data}}(\tilde{\mathbf{x}}) p_{\text{lat}}(\mathbf{z}), \quad (\text{A.5})$$

$$D_{ij} = \frac{1}{\sigma^2} \left[\sum_{k=1}^N (\mathbf{U}^{-1})_{ik} \xi_k \tilde{x}_k \right] z_j \tilde{p}_{\text{data}}(\tilde{\mathbf{x}}) p_{\text{lat}}(\mathbf{z}), \quad (\text{A.6})$$

$$E_{ij} = \frac{1}{\sigma^2} \left[\sum_{k=1}^N (\mathbf{U}^{-1})_{ik} \xi_k \tilde{x}_k \right] (z_j^2 - 1) \tilde{p}_{\text{data}}(\tilde{\mathbf{x}}) p_{\text{lat}}(\mathbf{z}). \quad (\text{A.7})$$

Here, we perform the transformation (PCA whitening [27])

$$x_i \rightarrow \tilde{x}_i = \sum_{j=1}^N U_{ij} (x_j - \langle x_j \rangle) / \xi_i, \quad (\text{A.8})$$

and correspondingly,

$$p_{\text{data}}(\mathbf{x}) \rightarrow \tilde{p}_{\text{data}}(\tilde{\mathbf{x}}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\tilde{x}_i^2}{2}\right). \quad (\text{A.9})$$

The orthogonal matrix $\mathbf{U}$ ($\mathbf{U}^{-1} = \mathbf{U}^\top$) transforms the data $\mathbf{x}$ into the directions of its principal components (eigenvalues of the covariance matrix) ξ_i^2 ($i = 1, 2, \dots, N$). The trivial extrema point is stable if $\delta^2 \mathcal{L}_{\text{VAE}}^G < 0$.

We expand $\delta f(\mathbf{z})$, $\delta \mu(\tilde{\mathbf{x}})$ and $\delta \sigma(\tilde{\mathbf{x}})$ in terms of Hermite polynomials H_n :

$$\delta f_i(\mathbf{z}) = \sum_{m_1, \dots, m_n} \alpha_{m_1, \dots, m_n}^i H_{m_1}(z_1) \cdots H_{m_n}(z_n), \quad (\text{A.10})$$

$$\delta \mu_j(\tilde{\mathbf{x}}) = \sum_{m_1, \dots, m_N} \beta_{m_1, \dots, m_N}^j H_{m_1}(\tilde{x}_1) \cdots H_{m_N}(\tilde{x}_N), \quad (\text{A.11})$$

$$\delta \sigma_j(\tilde{\mathbf{x}}) = \sum_{m_1, \dots, m_N} \gamma_{m_1, \dots, m_N}^j H_{m_1}(\tilde{x}_1) \cdots H_{m_N}(\tilde{x}_N). \quad (\text{A.12})$$

We choose Hermite polynomials

$$H_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}, \quad (\text{A.13})$$

due to their orthogonality properties [28]

$$\int H_m(x) H_n(x) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = n! \delta_{mn}, \quad (\text{A.14})$$

and the recurrence relations [28]

$$xH_n(x) = H_{n+1}(x) + nH_{n-1}(x), \quad (\text{A.15})$$

$$x^2 H_n(x) = H_{n+2}(x) + (2n+1)H_n(x) + n(n-1)H_{n-2}(x). \quad (\text{A.16})$$

Then, we can write each part of the integral in Eq. (A.1) as:

$$\int A_{ij} \delta f_i \delta f_j d\tilde{\mathbf{x}} dz = -\frac{1}{\sigma'^2} m_1! \cdots m_n! (\alpha_{m_1, \dots, m_n}^i)^2, \quad (\text{A.17})$$

$$\int B_{jk} \delta \mu_j \delta \mu_k d\tilde{\mathbf{x}} dz = -m_1! \cdots m_N! (\beta_{m_1, \dots, m_N}^j)^2, \quad (\text{A.18})$$

$$\int C_{jk} \delta \sigma_j \delta \sigma_k d\tilde{\mathbf{x}} dz = -2m_1! \cdots m_N! (\gamma_{m_1, \dots, m_N}^j)^2, \quad (\text{A.19})$$

$$\int D_{ij} \delta f_i \delta \mu_j d\tilde{\mathbf{x}} dz = \frac{1}{\sigma'^2} \left[\sum_{k=1}^N (U^{-1})_{ik} \xi_k \right] \alpha_{m_j=1}^i \beta_{m_k=1}^j, \quad (\text{A.20})$$

$$\int E_{ij} \delta f_i \delta \mu_j d\tilde{\mathbf{x}} dz = \frac{2}{\sigma'^2} \left[\sum_{k=1}^N (U^{-1})_{ik} \xi_k \right] \alpha_{m_j=2}^i \gamma_{m_k=1}^j, \quad (\text{A.21})$$

and calculate $\delta^2 \mathcal{L}_{\text{VAE}}^G$. Here, we use $\alpha_{m_j=1}^i$ to denote $\alpha_{m_1, \dots, m_n}^i$ if $m_j = 1$ and all other $m_i = 0$ for $i \neq j$. The same convention applies to $\alpha_{m_j=2}^i$, $\beta_{m_i=1}^j$ and $\gamma_{m_i=1}^j$. Thus, ensuring the negative definiteness of $\delta^2 \mathcal{L}_{\text{VAE}}^G$ reduces to checking the negative definiteness of the following quadratic expressions:

$$-\frac{1}{\sigma'^2} \sum_{i=1}^N \sum_{j=1}^n (\alpha_{m_j=1}^i)^2 - \sum_{i=1}^N \sum_{j=1}^n (\beta_{m_i=1}^j)^2 + \frac{2}{\sigma'^2} \sum_{i,k=1}^N \sum_{j=1}^n (U^{-1})_{ik} \xi_k \alpha_{m_j=1}^i \beta_{m_k=1}^j, \quad (\text{A.22})$$

$$-\frac{2}{\sigma'^2} \sum_{i=1}^N \sum_{j=1}^n (\alpha_{m_j=2}^i)^2 - 2 \sum_{i=1}^N \sum_{j=1}^n (\gamma_{m_i=1}^j)^2 + \frac{4}{\sigma'^2} \sum_{i,k=1}^N \sum_{j=1}^n (U^{-1})_{ik} \xi_k \alpha_{m_j=2}^i \gamma_{m_k=1}^j, \quad (\text{A.23})$$

both of which lead to the same condition.

Take the quadratic expression (A.22) for example, to derive the condition for negative definiteness, we denote vector α^* and β^* as

$$\alpha_{i+(j-1)n}^* \equiv \alpha_{m_j=1}^i, \quad (\text{A.24})$$

$$\beta_{i+(j-1)n}^* \equiv \beta_{m_i=1}^j. \quad (\text{A.25})$$

We also denote $\Lambda = \text{diag}(\xi_1, \dots, \xi_N)$. Further, we can write the quadratic expression (A.22) as a matrix form

$$-(\alpha^{*\top}, \beta^{*\top}) M \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad (\text{A.26})$$

where

$$M \equiv \sigma'^{-2} \begin{pmatrix} I_{Nn} & -[I_n \otimes (U^{-1} \Lambda)] \\ -[I_n \otimes (U^{-1} \Lambda)]^\top & \sigma'^2 I_{Nn} \end{pmatrix}, \quad (\text{A.27})$$

I is the identity matrix, and $\otimes$ refers to the Kronecker product. As I_{Nn} is positive definite, the positive definite of its Schur complement [29] S :

$$\begin{aligned}
S &= \sigma'^2 I_{Nn} - \left[I_n \otimes (U^{-1} \Lambda) \right]^\top \left[I_n \otimes (U^{-1} \Lambda) \right] \\
&= \sigma'^2 I_{Nn} - \left[I_n \otimes (\Lambda^\top (U^{-1})^\top U^{-1} \Lambda) \right] \\
&= \sigma'^2 I_{Nn} - \left[I_n \otimes (\Lambda^\top \Lambda) \right] \\
&= I_n \otimes \text{diag}(\sigma'^2 - \xi_1^2, \dots, \sigma'^2 - \xi_N^2),
\end{aligned} \tag{A.28}$$

leads to the positive definiteness of M and the negative definiteness of the quadratic expression (A.22), which requires the criterion (26):

$$\sigma'^2 > \max[\xi_1^2, \dots, \xi_N^2]. \tag{A.29}$$

The criterion is consistent with results obtained for linear VAEs [11, 12, 13]. Binomials (A.22) and (A.23) requires the same condition to be negative definiteness. $\alpha_{m_j=1}^i$ and $\beta_{m_i=1}^j$ in binomials (A.22) describes the linear perturbation (H_1 is linear) of $f_i(\tilde{z})$ and $\mu_j(\tilde{x})$ from the trivial extrema point respectively, which corresponds to the linear VAE setting where $f(z) = Vz + b$ and $\mu(x) = Wx + c$ [11, 12, 13].

References

- [1] D. P. Kingma, M. Welling, et al., [Auto-encoding variational bayes](#) (2013).
URL <https://arxiv.org/abs/1312.6114>
- [2] L. Girin, S. Leglaive, X. Bie, J. Diard, T. Hueber, X. Alameda-Pineda, [Dynamical variational autoencoders: A comprehensive review](#), arXiv preprint arXiv:2008.12595 (2020).
URL <https://arxiv.org/abs/2008.12595>
- [3] D. M. Blei, A. Kucukelbir, J. D. McAuliffe, [Variational inference: A review for statisticians](#), Journal of the American Statistical Association 112 (518) (2017) 859–877.
URL <https://www.tandfonline.com/doi/full/10.1080/01621459.2017.1285773>
- [4] M. I. Jordan, Z. Ghahramani, T. S. Jaakkola, L. K. Saul, [An introduction to variational methods for graphical models](#), Machine Learning 37 (1999) 183–233.
URL <https://link.springer.com/article/10.1023/A:1007665907178>
- [5] H. Huang, R. He, Z. Sun, T. Tan, et al., [Introvae: Introspective variational autoencoders for photographic image synthesis](#), Advances in Neural Information Processing Systems 31 (2018).
URL <https://proceedings.neurips.cc/paper/2018/hash/093f65e080a295f8076b1c5722a46aa2-Abstract.html>
- [6] S. Lin, R. Clark, R. Birke, S. Schönborn, N. Trigoni, S. Roberts, [Anomaly detection for time series using vae-lstm hybrid model](#), in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 4322–4326.
URL <https://ieeexplore.ieee.org/abstract/document/9053558>
- [7] O. Dollar, N. Joshi, J. Pfandtner, D. A. Beck, [Efficient 3d molecular design with an e \(3\) invariant transformer vae](#), The Journal of Physical Chemistry A 127 (37) (2023) 7844–7852.
URL <https://pubs.acs.org/doi/full/10.1021/acs.jpca.3c04188>
- [8] M. Rolinek, D. Zietlow, G. Martius, [Variational autoencoders pursue pca directions \(by accident\)](#), in: Proceedings of the IEEE/CVF Conference on Computer Cision and Pattern Recognition, 2019, pp. 12406–12415.
URL https://openaccess.thecvf.com/content_CVPR_2019/html/Rolinek_Variational_Autoencoders_Pursue_
- [9] A. Asperti, [Variance loss in variational autoencoders](#), in: International Conference on Machine Learning, Optimization, and Data Science, Springer, 2020, pp. 297–308.
URL https://link.springer.com/chapter/10.1007/978-3-030-64583-0_28

- [10] A. Nakagawa, K. Kato, T. Suzuki, [Quantitative understanding of vae as a non-linearly scaled isometric embedding](#), in: International Conference on Machine Learning, PMLR, 2021, pp. 7916–7926.
URL <https://proceedings.mlr.press/v139/nakagawa21a.html>
- [11] J. Lucas, G. Tucker, R. B. Grosse, M. Norouzi, [Don’t blame the elbo! a linear vae perspective on posterior collapse](#), Advances in Neural Information Processing Systems 32 (2019).
URL <https://proceedings.neurips.cc/paper/2019/hash/7e3315fe390974fcf25e44a9445bd821-Abstract.html>
- [12] H. Dang, T. Tran, T. Nguyen, N. Ho, [Beyond vanilla variational autoencoders: Detecting posterior collapse in conditional and hierarchical models](#), arXiv preprint arXiv:2306.05023 (2023).
URL <https://arxiv.org/abs/2306.05023>
- [13] Z. Wang, Z. Liu, [Posterior collapse of a linear latent variable model](#), Advances in Neural Information Processing Systems 35 (2022) 37537–37548.
URL https://openreview.net/forum?id=zAc2a6_0aHb
- [14] B. Dai, Z. Wang, D. Wipf, [The usual suspects? reassessing blame for vae posterior collapse](#), in: International Conference on Machine Learning, PMLR, 2020, pp. 2313–2322.
URL <https://proceedings.mlr.press/v119/dai20c.html>
- [15] S. R. Bowman, L. Vilnis, O. Vinyals, A. M. Dai, R. Jozefowicz, S. Bengio, [Generating sentences from a continuous space](#), arXiv preprint arXiv:1511.06349 (2015).
URL <https://arxiv.org/abs/1511.06349>
- [16] D. P. Kingma, T. Salimans, R. Jozefowicz, X. Chen, I. Sutskever, M. Welling, [Improved variational inference with inverse autoregressive flow](#), Advances in Neural Information Processing Systems 29 (2016).
URL https://proceedings.neurips.cc/paper_files/paper/2016/hash/ddeebdeefdb7e7e7a697e1c3e3d8ef54-A.html
- [17] C. K. Sønderby, T. Raiko, L. Maaløe, S. K. Sønderby, O. Winther, [Ladder variational autoencoders](#), Advances in Neural Information Processing Systems 29 (2016).
URL <https://proceedings.neurips.cc/paper/2016/hash/6ae07dcb33ec3b7c814df797cbda0f87-Abstract.html>
- [18] C.-W. Huang, S. Tan, A. Lacoste, A. C. Courville, [Improving explorability in variational inference with annealed variational objectives](#), Advances in Neural Information Processing Systems 31 (2018).
URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/65b0df23fd2d449ae1e4b2d27151d73b-A.html
- [19] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, A. Lerchner, [beta-vae: Learning basic visual concepts with a constrained variational framework](#), in: International Conference on Learning Representations, 2017.
URL <https://openreview.net/forum?id=Sy2fzU9g1>
- [20] M. D. Hoffman, M. J. Johnson, [Elbo surgery: yet another way to carve up the variational evidence lower bound](#), in: Workshop in Advances in Approximate Bayesian Inference, NIPS, Vol. 1, 2016.
- [21] Z. Liu, M. Ueda, [Zeroth, first, and second-order phase transitions in deep neural networks](#), Phys. Rev. Research 5 (4) (2023) 043243.
URL <https://journals.aps.org/prresearch/abstract/10.1103/PhysRevResearch.5.043243>
- [22] C. Cremer, X. Li, D. Duvenaud, [Inference suboptimality in variational autoencoders](#), in: International Conference on Machine Learning, PMLR, 2018, pp. 1078–1086.
URL <https://proceedings.mlr.press/v80/cremer18a.html>
- [23] D. J. Rezende, F. Viola, [Taming vases](#), arXiv preprint arXiv:1810.00597 (2018).
URL <https://arxiv.org/abs/1810.00597>

- [24] M. E. Tipping, C. M. Bishop, [Probabilistic principal component analysis](#), Journal of the Royal Statistical Society Series B: Statistical Methodology 61 (3) (1999) 611–622.
URL <https://academic.oup.com/jrsssb/article-abstract/61/3/611/7083217>
- [25] D. Kunin, J. Bloom, A. Goeva, C. Seed, [Loss landscapes of regularized linear autoencoders](#), in: International Conference on Machine Learning, PMLR, 2019, pp. 3560–3569.
URL <https://proceedings.mlr.press/v97/kunin19a.html>
- [26] A. Krizhevsky, G. Hinton, et al., [Learning multiple layers of features from tiny images](#) (2009).
URL <https://www.cs.toronto.edu/~kriz/cifar.html>
- [27] A. Kessy, A. Lewin, K. Strimmer, [Optimal whitening and decorrelation](#), The American Statistician 72 (4) (2018) 309–314.
URL <https://www.tandfonline.com/doi/full/10.1080/00031305.2016.1277159>
- [28] L. C. Fai, Special Functions in Physics and Engineering: A renewed approach with applications, IOP Publishing, 2024.
- [29] F. Zhang, The Schur complement and its applications, Vol. 4, Springer Science & Business Media, 2006.